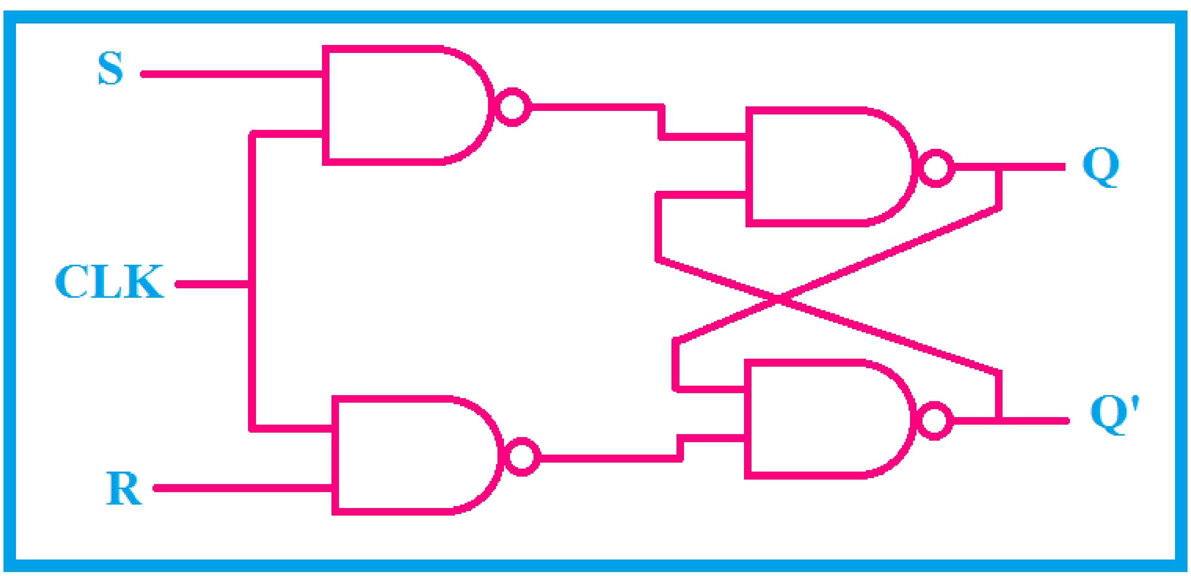
Day 26

SR Flip-Flop (Edge-Triggered) in Verilog

Circuit Diagram :

S	R	Q_{n+1}
0	0	Q_n(No Change)
0	1	0
1	0	1
1	1	x



Verilog Code

SR-FlipFlop verilog code

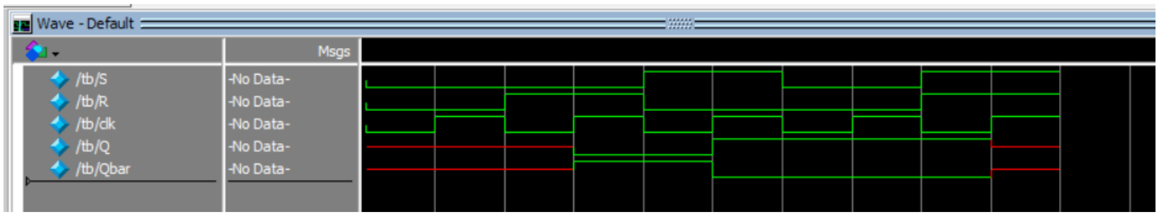
```
module SR_FF(S,R,clk,Q,Qbar);
input S,R,clk;
output reg Q;
output Qbar;
always @(posedge clk) begin
case({S,R})
2'b00 : Q<= Q;
2'b01 : Q<= 1'b0;
2'b10 : Q<= 1'b1;
2'b11 : Q<= 1'bx;
endcase
end
assign Qbar=~Q;
endmodule
```

Testbench Code

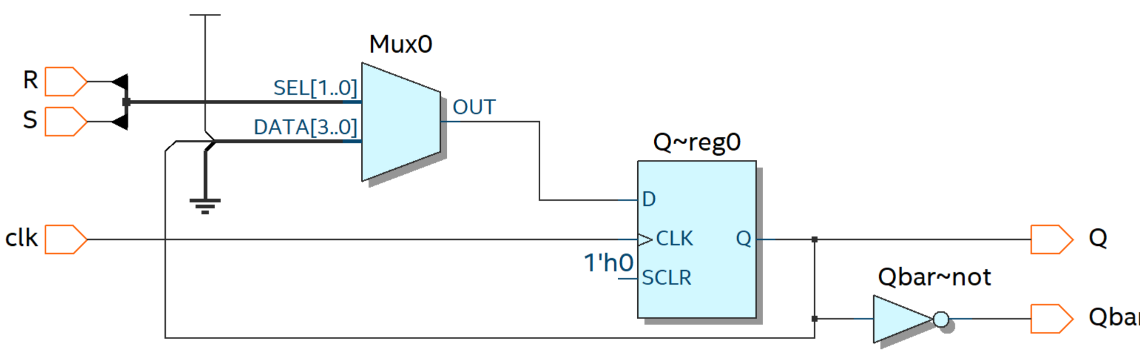
SR-FlipFlop Tb code

```
module tb;
reg S, R, clk;
wire Q, Qbar;
SR_FF dut (S,R,clk,Q,Qbar);
initial begin
clk = 0;
forever #5 clk = ~clk;
end
initial begin
$monitor("Time=%0t S=%b R=%b Q=%b Qbar=%b",$time,S,R,Q,Qbar);
S=0; R=0; #10
S=0; R=1; #10
S=1; R=0; #10
S=0; R=0; #10
S=1; R=1; #10
$finish;
end
endmodule
```

Waveform



Schematic



EDA Tools Used

- IntelQuartusPrime
- ModelSim